Lecture 31

4/8/26
JRA

Computed Tomography (CT)

Single projection leads to superposition

CT - take 1000s of slices

See BB explanation & slide w/ cylinder, ...

Original incarnation

Pencil beam + single detector — 160
Translate / rotate / translate / ... — 180 angles

Did math, reconstructed attenuation coefficients as a fcn of position

SLOW

Now on 3rd generation CT scanner (4th fizzled)
Rotate / rotate scanner
5 min → seconds

Key attributes ① fan beam of X-rays that encompasses entire patient width (so no need to translate)

(2) Use many detectors
↳ Source & detectors rotate in synchrony to generate different views

(3) To get different slices thru patient

 → z

see SSCT vs MSCT
↑ single slice — multislice

SSCT Original scanners detectors were <u>NOT</u> divided along z ⇒ one slice at a time detected

Now have multi-slice CT scanners divided along z (up to 64 simultaneous slices)

intermittent (4) Patient movement along z used to be stop start (stop collect start)

continuous Now continuous movement called spiral CT

Collecting <u>LOTS</u> of data
How do you reconstruct slices?

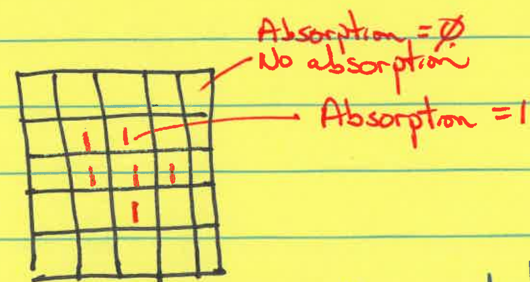
Reconstruction

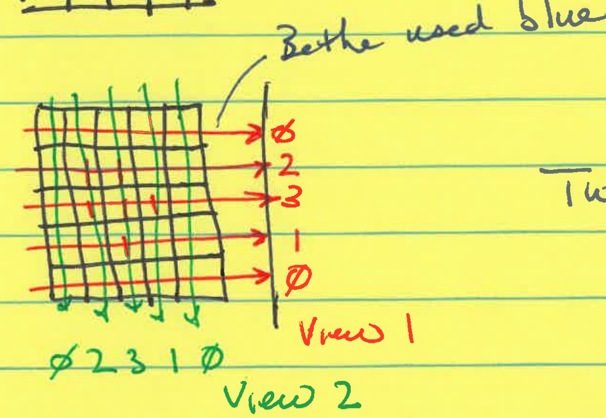
Current gold std is filtered back projection (FBP)

Start w/ simplified version

Start w/ simple back projection (SBP)

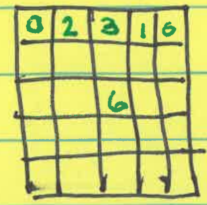
Doctor makes decision about resolution





Two views

Back projection — smear back & add attenuations



Bulk of attenuating material is where it's supposed to be: central

Looked at PP slide w/ phantoms

Smear & add

smear forward to get projection
smear back & add

(1) Bulk of atten material in the correct location
(2) Blur exists (plagued by blur)

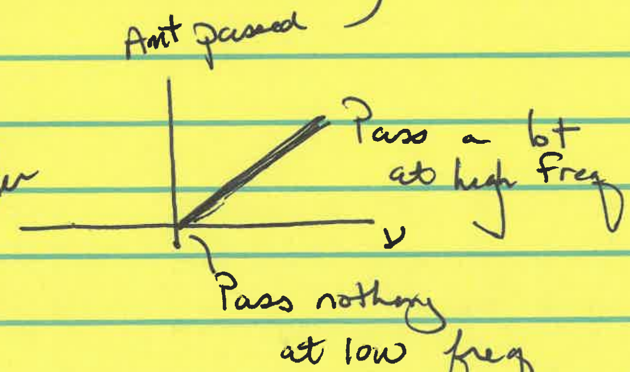
Filtered back projection eliminates blur

Is blur a low-frequency or high-frequency?
coarse details / fine details

Blur is excess low-frequency information in the image, so want a filter that removes low-freq info

Most rudimentary filter is the ramp filter

Ramp filter



SUMMARY

Strengths - Fast, cheap
Excellent images of skeleton
Excellent soft tissue contrast using CT
(or contrast agent)
Anyone can have a CT scan
but not an MRI
Less sensitive to movement

Weaknesses Ionizing radiation →
μ NOT the most sensitive measure
of fcn
Better ways to get anatomical
information

MOVIES

COVID movie w/ lung damage

Great anatomical technique

Nuclear Medicine Imaging

Radionuclide Imaging

Often used w/
X-ray techniques

X-Ray	Nuclear
(1) Transmission-based (X-rays transmitted thru body)	(1) Emission-based (gamma rays emitted from body)
(2) Anatomical info	(2) Functional info
(3) Excellent resolution (0.1 mm → 1 mm) NOT diffraction limit	(3) Relatively poor res 5 mm - 3 cm
(4) Poorer contrast	(4) Excellent intrinsic contrast (bkgrd ≈ zero) signal against dk bkgd
(5) Less sensitive	☆ (5) Extremely sensitive

Doses in nuclear medicine are very low, so

Sensitivity is excellent
- Required doses extremely low
- Prob of perturbing fcn w/ what you add is correspondingly low

→ Good way to assay fcn

Contrast agents in X-ray & MRI may cause issues

simultaneous - A + F

Name JRA

Physics 390 In-Class Activity 11

(DO NOT INCLUDE SOURCES WHEN YOU BACK PROJECT)

OBJECT

					P1
0	0	0	0	0	Ø
0	0	0	0	0	Ø
0	0	12	0	0	12
0	1	2	1	0	4
0	0	0	0	0	Ø
Ø	1	14	1	Ø	P2

RECONSTRUCTION

					P1
0 0 Ø	0 1 1	0 14 14	0 1 1	0 Ø	Ø
0 0 Ø	0 1 1	0 14 14	0 1 1	0 0 Ø	Ø
12 0 12	12 1 13	12 14 26	12 1 13	12 0 12	12
4 0 4	4 1 5	4 14 18	4 1 5	4 0 4	4
0 0 Ø	0 1 1	0 14 14	0 1 1	0 0 Ø	Ø
Ø	1	14	1	Ø	P2

Red # = Blue # + Green #

Questions

1. The bulk of the attenuating material is in the center of the object. Is this also true for the image produced by this very rudimentary simple back projection?

Yes - the bulk of the attenuating material is in the center of the reconstruction

2. Does this back projection suffer from blur (i.e., attenuating material appearing in incorrect parts of the image)?

Yes

3. Do you think that medicine can do better?

Yes